Supplementary File

Search items for HINARI/PubMed

#1= "Prevalence"[MeSH] OR "Epidemiology"[MeSH]

#2= "Malnutrition"[MeSH] OR "Obesity"[MeSH]

#3= malnutrition [tiab] OR nutrition*[tiab] OR obesity [tiab]

#4= aged [MeSH] OR elder [MeSH]

#5= old*[tiab] OR aged [tiab] OR elder [tiab] OR geriatric [tiab]

#6= Africa [MeSH] OR Africa [tiab]

#7= Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion[tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab]

#8=#2 OR #3

#9= #4 OR #5

#10= #6 OR #7

#11= #1 AND #8 AND #9 AND #10

#12= Filter: date from 2000/01/01 to 2019/12/10; English language; Humans

Combined search terms

((("Prevalence"[MeSH] OR "Epidemiology"[MeSH]) AND ("Malnutrition"[MeSH] OR "Obesity"[MeSH] OR malnutrition [tiab] OR nutrition*[tiab] OR obesity[tiab]) AND (aged[MeSH] OR elder[MeSH] OR old*[tiab] OR aged[tiab] OR elder[tiab] OR geriatric[tiab]) AND (Africa[MeSH] OR Africa[tiab] OR Angola[tiab] OR Benin[tiab] OR Botswana[tiab] OR Burkina Faso[tiab] OR Burundi[tiab] OR Cameroon[tiab] OR Cape Verde[tiab] OR Central

African Republic[tiab] OR Chad[tiab] OR Comoros[tiab] OR Congo[tiab] OR Côte d'Ivoire[tiab] OR Democratic Republic of the Congo[tiab] OR Djibouti[tiab] OR Equatorial Guinea[tiab] OR Eritrea[tiab] OR Ethiopia[tiab] OR Gabon[tiab] OR Gambia[tiab] OR Ghana[tiab] OR Guinea[tiab] OR Guinea-Bissau[tiab] OR Kenya[tiab] OR Lesotho[tiab] OR Liberia[tiab] OR Madagascar[tiab] OR Malawi[tiab] OR Mali[tiab] OR Mauritania[tiab] OR Mauritius[tiab] OR Mayotte[tiab] OR Mozambique[tiab] OR Namibia[tiab] OR Niger[tiab] OR Nigeria[tiab] OR Réunion [tiab] OR Rwanda[tiab] OR Saint Helena[tiab] OR São Tomé and Príncipe[tiab] OR Senegal[tiab] OR Seychelles[tiab] OR Sierra Leone[tiab] OR Somalia[tiab] OR South Africa[tiab] OR Sudan[tiab] OR Swaziland[tiab] OR Togo[tiab] OR Uganda[tiab] OR United Republic of Tanzania[tiab] OR Zambia[tiab] OR Zimbabwe[tiab] OR Algeria[tiab] OR Egypt[tiab] OR Libya[tiab] OR Morocco[tiab] OR Tunisia[tiab])))

In case of Google Scholar advanced search for the topic was as follows:

1. With all of the words: old age people, or elderly
2. With at least one of the words: prevalence, epidemiology, malnutrition, undernutrition, overnutrition, obesity, overweight, old age people, older adult, elderly, Africa, Sub-Saharan Africa, Horn of Africa
3. Where my words occur: in the title of the article

Supplementary Table 1: Quality appraisal checklist for prevalence studies

Items		**Quality score**
	External validity Criteria	
1.	Was the study's target population a close representation of the national population in relation to relevant variables?	(1 point)
2.	Was the sampling frame a true or close representation of the target population?	(1 point)
3.	Was some form of random selection used to select the sample, OR was census undertaken?	(1 point)
4.	Was the likelihood of nonresponse bias minimal?	(1 point)
		Total (4 points)
	Internal validity Criteria	
5.	Were data collected directly from the subjects (as opposed to a proxy)?	(1 point)
6.	Was an acceptable case definition used in the study?	(1 point)
7.	Was the study instrument that measured the parameter of interest shown to have validity and reliability?	(1 point)
8.	Was the same mode of data collection used for all subjects?	(1 point)

9.	Was the length of the shortest prevalence period for the parameter of interest appropriate?	(1 point)
10.	Were the numerator(s) and denominator(s) for the parameter of interest appropriate?	(1 point)
		Total (6 points)
	Overall total	10 points

Methodological quality

High quality: score > 8

Moderate quality: score 6-8

Low quality: score 0-5

(Source: Hoy et al.(24))

Supplementary Table 2. Characteristics of Selected Studies Using Different Nutritional Assessment Tools

S/no	Authors	Country	Study design	Study setting	Sample size	Age group	Gender		Assessment tool used	Nutritional status (outcome variable) in %				Multi-morbidity
							Male	Female		Under-nutrition	At risk of malnutrition	Normal-weight/ Nutrition	Overweight/ obesity	
1.	Aganiba, B. et al., 2015 (62)	Ghana	C/S	Community	400	≥ 65	177	223	BMI	18.0		60.5	21.5	Not reported
2.	Apprey C. et al., 2019 (63)	Ghana	C/S	Community	375	≥ 60	127	248	BMI	9.9		48	30.9/11.2	Multi
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	Cohort	Hospital	222	≥ 60	121	101	BMI	25.2		55.0	14.9/5.0	One
4.	Mkhize X. et al., 2013 (53)	South Africa	C/S	Hospital	270	≥ 60	46	224	BMI	4.0		19.0	24.0/53.0	One
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	C/S	Community	537	≥ 60	227	310	BMI	26.4		58.3	10.8/4.5	Not reported
6.	Ijarotimi & Keshinro, 2004 (65)	Tanzania	C/S	Hospital	100	≥ 60	51	49	BMI	22.0		61.0	17.0	Not reported
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	C/S	Hospital	121	≥ 60	67	54	BMI	25.6		74.4	0	Not reported
8.	Olayiwola O.et al, 2006 (47)	Nigeria	C/S	Hospital	295	≥ 60	153	142	BMI	18.7		60.7	15.7/4.9	Not reported
9.	Jésus p. et al., 2017 (60)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
10.	Rouvray C. et al., 2014 (61)	CAR & ROC	C/S	Community	990	≥ 65	405	585	BMI	19.2		52.9	19.1/8.8	Multi
11.	Legesse M. et al., 2019 (34)	Ethiopia	C/S	Community	892	≥ 65	365	527	BMI	17.6		82.4		Multi
12.	Wondiye K et al., 2019 (35)	Ethiopia	C/S	Community	554	≥ 60	295	259	BMI	17.1		80.6	2.3	Not reported
13.	Agbozo, F. et al., 2018 (64)	Ghana	C/S	Community	120	60-70	42	78	BMI	10.0		51.7	21.7/16.6	Not reported
14.	Alao M. et al., 2015 (48)	Nigeria	Case-control	LTC	418	≥ 60	191	227	BMI	6.9		53.3	24.6/15.1	Not reported
15.	Tessfamichael D. et al., 2014 (40)	Ethiopia	C/S	Community	757	≥ 65	194	563	BMI	21.9		78.1		Multi
16.	Otitoola Oc. Et al., 2015 (57)	South Africa	Cohort	Elderly day care center	46	≥ 60	8	38	BMI	0		18.4	80.4	Not reported
17.	Pilleron, S. et al., 2015 (58)	CAR & ROC	C/S	Community	645	> 65	Not reported	Not reported	BMI	68.5		31.5		Not reported
18.	Esmayel EM. et al, 2013 (42)	Egypt	C/S	Community	200	≥65	112	88	NCL	56.0	18.0	26.0		Not reported
19.	Andre MB. et al., 2013 (59)	ROC	C/S	Community	370	≥60	194	176	MNA	13.8	28.4	57.8		Multi
20.	Adebusoye L. A. et al., 2012 (49)	Nigeria	C/S	Hospital	500	≥60	176	324	MNA	7.8	11.8	80.4		Multi
21.	Robb L. et al., 2017 (54)	South Africa	C/S	LTC	124	≥60	35	89	MNA	7.3	55.6	37.1		Not reported
22.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	C/S	Community	2,219	≥60	1165	1054	MNA-SF	10.9	41.9	47.2		Multi

23.	Naidoo I. et al., 2015 (56)	South Africa	C/S	Community	984	≥60	224	760	MNA SF	51.0	5.5	43.4		Multi
24.	Adhana, Z K et al., 2015 (37)	Ethiopia	C/S	Community	423	≥60	99	324	MNA-SF	77.3	0	22.7		Multi
25.	Adebusoye L. et al., 2019 (50)	Nigeria	C/S	Hospital	350	>60	Not reported	Not reported	MNA-SF	4.57	95.43			Multi
26.	Marias ML. et al., 2007 (55)	South Africa	C/S	Community	210	≥60	53	157	MNA	5.7	47.1	47.1		Not reported
27.	Hailemariam H. et al., 2016 (36)	Ethiopia	C/S	Community	548	≥60	261	287	MNA	9.3	28.3	62.4		Not reported
28.	Abate et al., 2020 (38)	Ethiopia	C/S	Community	662	≥65	240	422	MNA	26.6	73.4			One
29.	Abdu AO et al., 2020 (39)	Ethiopia	C/S	Community	592	≥65	291	301	MNA	15.7	51.7	32.6		One
30.	Abd Allah et al., 2020 (43)	Egypt	C/S	Hospital	194	≥60	107	87	MNA	33.0	51.5	15.5		Multi
31.	Rasha El-Desouky 2017 (44)	Egypt	C/S	Community	320	≥60	167	153	MNA	35.0	38.4	26.6		Multi
32.	Khater MS et al., 2011 (45)	Egypt	C/S	LTC	120	≥60	56	64	MNA	10.8	40.8	48.4		Multi
33.	F. Mahjoub et al., 2019 (68)	Tunis	C/S	Hospital	40	≥65	21	19	MNA	40.0	52.5	7.5		One
34.	Adebusoye, LA. et al., 2018 (52)	Nigeria	C/S	Hospital	624	≥60	246	378	MNA	2.24	32.69	65.06		One
35.	Abd-El-Gawad et al., 2014 (51)	Nigeria	Cohort	Hospital	131	≥60	67	64	MNA	40.46	52.67	6.87		Multi
36.	Mahfouz, E.M. et al., 2013 (46)	Egypt	C/S	Community	350	≥60	136	214	MNA	8.6	29.7	61.7		Multi
37.	Andia A, et al., 2019 (70)	Niger	C/S	Community	384	≥60	132	252	MNA	7.8	58.5	33.7		Multi

Abbreviations: BMI = Body mass index
C/S: Cross-section
LTC: Long term care
CAR = Central Africa Republic
MNA: Mini Nutritional Assessment Tool
MNA-SF: Mini Nutritional Assessment Tool- short Form
ROC= Republic of Congo
MUAC: mid-upper arm circumference
NCL: Nutritional Screening Checklist

Supplementary Table 3. Summary of the quality of included studies

S/no	Author & year of publication	Country	External validity quality Score				Internal validity quality Score						Total Score	Study Quality
			1	2	3	4	5	6	7	8	9	10		
1.	Aganiba, B. et al., 2015 (62)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
2.	Apprey C. et al., 2019 (63)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
3.	Diendéré, J. et al, 2018 (35)	Burkina Faso	0	1	1	1	1	1	1	1	1	1	9	High
4.	Mkhize X. et al., 2013 (53)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
5.	MJ, C. et al, 2012 (69)	Lake Victoria Basin of East Africa	0	1	1	1	1	1	1	1	1	1	9	High
6.	Ijarotimi OS & Keshinro OO, 2004 (65)	Tanzania	0	0	0	1	1	1	1	1	1	1	7	Moderate
7.	Nyaruhucha CNM. et al, 2001 (66)	Tanzania	0	1	1	1	1	1	1	1	1	1	9	High
8.	Olayiwola O.et al, 2006 (47)	Nigria	0	1	1	1	1	1	1	1	1	1	9	High
9.	Rouvray C. et al., 2014 (61)	Central Africa (COR & ROC)	0	1	1	1	1	1	1	1	1	1	9	High
10.	Legesse M. et al., 2019 (34)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
11.	Wondiye K et al., 2019 (35)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
12.	Agbozo, F. et al., 2018 (64)	Ghana	0	1	1	1	1	1	1	1	1	1	9	High
13.	Alao M. et al., 2015 (48)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate
14.	Tessfamichael, D. et al., 2014 (40)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
15.	Otitoola Oc. Et al., 2015 (57)	South Africa	0	0	0	0	1	1	1	1	1	1	6	Moderate
16.	Pilleron, S. et al., 2015 (58)	Central Africa (COR & ROC)	1	1	1	1	1	1	1	1	1	1	10	High

17.	Esmayel EM. et al, 2013 (42)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
18.	Andre MB. et al., 2013 (59)	Democratic Republic of Congo	0	1	1	1	1	1	1	1	1	1	9	High
19.	Adebusoye L.A. et al., 2012 (49)	Nigeria	0	1	1	1	1	1	1	1	1	1	9	High
20.	Robb L. et al., 2017 (54)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
21.	El-Sherbiny N.A. et al., al.2016 (41)	Egypt	0	1	1	1	1	1	1	1	1	1	9	High
22.	Naidoo I. et al., 2015 (56)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
23.	Adhana, Z K et al., 2015 (37)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
24.	Adebusoye L. et al., 2019 (50)		0	0	1	1	1	1	1	1	1	1	8	Moderate
25.	Marias ML. et al., 2007 (55)	South Africa	0	1	1	1	1	1	1	1	1	1	9	High
26.	Hailemariam H. et al., 2016 (36)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
27.	Abate et al., 2020 (38)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
28.	Abdu AO et al., 2020 (39)	Ethiopia	0	1	1	1	1	1	1	1	1	1	9	High
29.	Abd Allah et al., 2020 (43)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
30.	Rasha El-Desouky_2017 (44)	Egypt	0	0	1	1	1	1	1	1	1	1	8	Moderate
31.	Khater MS et al., 2011 (45)	Egypt	0	0	0	0	1	1	1	1	1	1	6	Moderate
32.	F. Mahjoub et al., 2019 (68)	Tunis	0	0	0	1	1	1	1	1	1	1	7	Moderate
33.	Adebusoye, LA. et al., 2018 (52)	Nigeria	0	0	0	0	1	1	1	1	1	1	6	Moderate
34.	Abd-El-Gawad et al., 2014 (51)	Nigeria	0	0	0	1	1	1	1	1	1	1	7	Moderate

35.	Mahfouz, E.M. et al., 2013 (46)	Egypt	0	0	0	1	1	1	1	1	1	1	7	Moderate
36.	Andia A, et al., 2019 (70)	Niger	0	0	0	0	1	1	1	1	1	1	6	Moderate